# Saddle-Dominated Capacity of the LSE Dense Associative Memory

**Abstract**

The log-sum-exp (LSE) Dense Associative Memory stores an exponentially large set of patterns on the $N$-sphere with a kernel sharpness parameter we call $\beta_{\text{net}}$. Ramsauer et al. [9] obtained a limiting storage load $\alpha_c = 1/2$ at zero temperature by modelling the inter-pattern overlaps as Gaussian; their bound is independent of $\beta_{\text{net}}$. We revisit the calculation under the exact spherical overlap density. The dominant contribution to the noise floor moves from the upper edge of the overlap distribution to an interior saddle of the partition-sum integrand, with saddle value $g_{\max}(\beta_{\text{net}})$. This single quantity controls every corrected boundary in the $(\alpha, T)$ plane. At $T = 0$ the corrected capacity is $\beta_{\text{net}} - g_{\max}(\beta_{\text{net}})$, the load at which the cusp of the leading-order free energy reaches the unit sphere and the retrieval basin disappears. It is finite, depends on $\beta_{\text{net}}$, sits below Ramsauer's bound at moderate $\beta_{\text{net}}$, and exceeds it at large $\beta_{\text{net}}$. Hybrid Monte Carlo at $N = 100$ places the empirical basin boundary on the corrected curve, consistent with the analytical prediction.

## 1 Introduction

The Hopfield network [3] stores patterns as minima of a pairwise energy and retrieves the closest stored pattern from a noisy cue. Classical pairwise storage saturates at a load of about 0.138 patterns per neuron [1]. Dense Associative Memory replaces the pairwise interaction by a strongly nonlinear energy and raises the capacity to an exponential $M = \mathrm{e}^{\alpha N}$ [4, 5, 2]. In the continuous version of the model, both the stored patterns $\boldsymbol{\xi}^\mu \in \mathbb{R}^N$ and the chain $\boldsymbol{x}$ live on the sphere $|\boldsymbol{x}|^2 = N$. The log-sum-exp (LSE) variant of Ramsauer et al. [9] writes

$$H(\boldsymbol{x}) = -\beta_{\text{net}}^{-1} \ln \sum_{\mu=1}^{M} \exp\!\big(\beta_{\text{net}}\, \boldsymbol{x} \cdot \boldsymbol{\xi}^\mu\big), \tag{1}$$

with $\beta_{\text{net}}$ a kernel sharpness parameter that scales the dot product inside the log-sum-exp. $\beta_{\text{net}}$ controls how strongly the energy distinguishes patterns and is independent of the simulation temperature $T$.

For continuous LSE, Ramsauer et al. [9] derived a limiting load $\alpha_c = 1/2$ at $T = 0$. The argument treats the inter-pattern overlap $\varphi_{1\mu} = \boldsymbol{\xi}^1 \cdot \boldsymbol{\xi}^\mu / N$ as a Gaussian with mean 0 and variance $1/N$, and asks at what load the typical maximum of $M$ such overlaps first reaches 1. The Gaussian extreme value satisfies $\varphi_{\max}^{(G)} = \sqrt{2\alpha}$, so the condition $\varphi_{\max}^{(G)} = 1$ gives $\alpha = 1/2$. The bound does not involve $\beta_{\text{net}}$.

The Gaussian is the leading term of the exact spherical overlap density near $\varphi = 0$, but the two densities diverge near $\varphi = 1$. Whether the corrected density alters the capacity was raised by Petrova et al. [7], who derived the finite-temperature retrieval boundary $T_c(\alpha)$ for LSE and LSR under the Gaussian approximation and flagged its noise-floor derivation as the candidate failure point. Petrova [6] subsequently measured this boundary by direct Monte Carlo at moderate $N$ for both kernels, without pursuing the low-$T$ limit near the limiting load. The compactly supported LSR variant was analysed by Petrova et al. [8] from a different angle: by direct Monte Carlo at $N \approx$ 40–50 they showed that the apparent retrieval boundary in the simulations is a kinetic crossover. A metastable strip sits below the thermodynamic boundary, separated from the competing-pattern basins by an entropic barrier whose Kramers escape time grows exponentially in $N$. They note that resolving the strip at larger $N$ would require a smarter Monte Carlo scheme than the direct method they employed, and do not provide one.

This paper revisits Ramsauer's $\alpha_c = 1/2$ for LSE under the exact spherical density, treating $\beta_{\text{net}}$ as a free parameter. The corrected capacity is finite and depends on $\beta_{\text{net}}$; it sits below Ramsauer's bound at moderate $\beta_{\text{net}}$, crosses it at $\beta_{\text{net}} \approx 0.7$, and grows logarithmically with $\beta_{\text{net}}$ beyond. Two boundary curves arise in the $(\alpha, T)$ plane: a single-pattern spinodal $\alpha_c^{\text{sp}}(\beta_{\text{net}}, T)$ at which the cusp of the leading-order free energy reaches the basin minimum and the basin disappears, and a thermodynamic crossover $\alpha_c^{\text{sd}}(\beta_{\text{net}}, T)$ at which the noise floor and the basin minimum exchange order in free energy. Both curves are set by the integrand saddle value $g_{\max}(\beta_{\text{net}})$, and they coincide at $T = 0$ at the endpoint $\beta_{\text{net}} - g_{\max}(\beta_{\text{net}})$. At $T > 0$ they separate; the spinodal is the relevant boundary for cold-initialised Monte Carlo, the saddle line is the thermodynamic crossover.

Two technical points motivate the rest of the paper. First, the naive correction — replacing the Gaussian by the exact density and applying the same right-edge logic — gives the wrong capacity: the LSE partition sum is governed by an interior saddle, not by the edge of the overlap distribution. Second, the local-refresh Monte Carlo schemes used to accelerate large-$N$ studies fail near the capacity boundary by introducing unquenched disorder; the simulations in this paper rely on a hybrid scheme that keeps the cusp-tail patterns explicit while treating the bulk analytically.

Section 2 develops the theory: the Gaussian baseline, the naive exact-density variant, the interior saddle, the cusp picture, the spinodal, and the two-protocol picture. Section 3 describes the three Monte Carlo schemes (direct, truncated, hybrid) used here, with a paragraph on why local refresh fails. Section 4 shows the empirical $N = 100$ heatmap together with the four boundary curves. Section 5 closes with the Ramsauer claim and the saddle-vs-spinodal distinction.

## 2 Theory

**Setup.** Both the stored patterns $\boldsymbol{\xi}^\mu$ and the chain $\boldsymbol{x}$ lie on the sphere $S^{N-1}(\sqrt{N})$. The inter-pattern overlap $\varphi_{1\mu} = \boldsymbol{\xi}^1 \cdot \boldsymbol{\xi}^\mu / N$ between a fixed retrieving pattern and a random spurious one is the projection of one uniform direction onto another; its density on $[-1, 1]$ is the spherical density

$$f(\varphi) = C_N \, (1 - \varphi^2)^{(N-3)/2}, \qquad C_N = \frac{\Gamma(N/2)}{\Gamma((N-1)/2)\sqrt{\pi}} \, . \tag{2}$$

At large $N$, $f$ concentrates near $\varphi = 0$ with variance $1/N$, but its right tail near $\varphi = 1$ decays as $(1-\varphi^2)^{N/2}$, lighter than any Gaussian of matched variance. The chain-target overlap $\varphi_1 = \boldsymbol{\xi}^1 \cdot \boldsymbol{x}/N$ measures retrieval.

For a chain in the basin of $\boldsymbol{\xi}^1$ the LSE log-sum is dominated by the $\mu = 1$ term, $H(\boldsymbol{x})/N \approx -\beta_{\text{net}}^{-1} \ln \mathrm{e}^{\beta_{\text{net}} N \varphi_1} = -\varphi_1$ — the $\beta_{\text{net}}$ factor inside the exponential and the $\beta_{\text{net}}^{-1}$ in front of the logarithm cancel for any single-term dominant contribution. Adding the perpendicular-subspace entropy $s(\varphi_1) = \frac{1}{2} \ln(1 - \varphi_1^2)$ and minimising $f = u - Ts$ with $u = 1 - \varphi_1$ gives the retrieval free energy per spin

$$f_{\text{ret}}(T) = (1 - \phi_{\text{eq}}) - \tfrac{T}{2} \ln(1 - \phi_{\text{eq}}^2), \qquad \phi_{\text{eq}}(T) = \tfrac{1}{2}\big(-T + \sqrt{T^2 + 4}\big), \tag{3}$$

$\beta_{\text{net}}$-independent by the same cancellation.

Away from the basin the noise floor comes from the $M - 1$ spurious sum

$$S(\boldsymbol{x}) \; = \; \sum_{\mu \neq 1} \mathrm{e}^{\beta_{\text{net}} N \varphi_\mu(\boldsymbol{x})}, \qquad \frac{1}{N} \ln S \; \approx \; \alpha + \frac{1}{N} \ln \int_{-1}^{1} f(\varphi) \, \mathrm{e}^{N \beta_{\text{net}} \varphi} \, \mathrm{d}\varphi, \tag{4}$$

where we used $M = \mathrm{e}^{\alpha N}$ and replaced the sum over patterns by an integral with density $f$ — valid in the bulk of $\boldsymbol{x}$ because the $\varphi_\mu$ are independent draws from $f$. The integrand exponent is $g(\varphi; \beta_{\text{net}}) = \frac{1}{2} \ln(1 - \varphi^2) + \beta_{\text{net}} \, \varphi$.

**Gaussian baseline.** Replacing $f$ by the moment-matched Gaussian $\sqrt{N/2\pi}\,\mathrm{e}^{-N\varphi^2/2}$ makes the exponent $-\frac{1}{2}\varphi^2+\beta_{\rm net}\varphi$ monotone in $\varphi$ on $[-1,1]$ for $\beta_{\rm net}\le 1$, so the integral (4) is set by the typical extreme of $M$ Gaussian samples, $\varphi_{\max}^{(G)}=\sqrt{2\alpha}$. Substituting gives $N^{-1}\ln S\to\beta_{\rm net}\sqrt{2\alpha}$ and a noise-floor energy per spin $u_{\rm noise}^{(G)}=1-\sqrt{2\alpha}$. The boundary $u_{\rm noise}^{(G)}=f_{\rm ret}(T)$ reads

$$\alpha_c^{\rm G}(T)=\tfrac{1}{2}\big[1-f_{\rm ret}(T)\big]^2,\qquad \alpha_c^{\rm G}(0)=\tfrac{1}{2}. \tag{5}$$

The $\beta_{\rm net}^{-1}$ prefactor of $H$ and the $\beta_{\rm net}$ inside the exponent cancel each other at the right edge, so (5) is $\beta_{\rm net}$-independent. This is Ramsauer's bound.

**Naive correction: exact density at the edge.** Keep the exact density $f$ but apply the same edge logic. The typical maximum of $M$ samples from $f$ satisfies $M\,\mathrm{Pr}_f(\varphi>\varphi_{\max})=1$, which after computing the right tail of (2) gives $\varphi_{\max}(\alpha)=\sqrt{1-\mathrm{e}^{-2\alpha}}$. The exponent at the edge is $g(\varphi_{\max})=-\alpha+\beta_{\rm net}\,\varphi_{\max}$, so $N^{-1}\ln S\to\beta_{\rm net}\,\varphi_{\max}(\alpha)$ and $u_{\rm noise}^{(\rm ex)}=1-\varphi_{\max}(\alpha)$. Solving $u_{\rm noise}^{(\rm ex)}=f_{\rm ret}(T)$ yields

$$\alpha_c^{\rm bd}(T)\;=\;-\tfrac{1}{2}\ln\!\big[1-(1-f_{\rm ret}(T))^2\big],\qquad \alpha_c^{\rm bd}(0)=+\infty. \tag{6}$$

The boundary diverges at $T=0$: the exact density has no finite right edge, so the edge logic returns no bound. This is the right answer for the compactly supported LSR kernel — there the kernel cuts the integral off at a finite $\varphi_c<1$ [8]. For LSE the edge logic is wrong whenever the integrand has an interior maximum inside $[0,1)$.

**Interior saddle: the correct boundary.** The exponent $g(\varphi;\beta_{\rm net})$ has derivative $g'(\varphi)=\beta_{\rm net}-\varphi/(1-\varphi^2)$, which vanishes at the positive root

$$\varphi^*(\beta_{\rm net})=\frac{-1+\sqrt{1+4\beta_{\rm net}^2}}{2\beta_{\rm net}}. \tag{7}$$

The saddle is inside $(0,1)$ for every $\beta_{\rm net}>0$. By Laplace's method,

$$\int_{-1}^{1} f(\varphi)\,\mathrm{e}^{N\beta_{\rm net}\varphi}\,\mathrm{d}\varphi\;\sim\;\sqrt{\frac{2\pi}{N\,|g''(\varphi^*)|}}\,\mathrm{e}^{N\,g_{\max}(\beta_{\rm net})},\qquad g_{\max}(\beta_{\rm net})\;\equiv\;g(\varphi^*;\beta_{\rm net})\;=\;\tfrac{1}{2}\ln(1-\varphi^{*\,2})+\beta_{\rm net}\,\varphi^*, \tag{8}$$

so $N^{-1}\ln S\;\to\;\alpha+g_{\max}(\beta_{\rm net})$ and the noise-floor energy per spin reads $u_{\rm noise}^{(\rm sd)}=1-[\alpha+g_{\max}(\beta_{\rm net})]/\beta_{\rm net}$. Equating this to $f_{\rm ret}(T)$ gives the saddle-dominated boundary

$$\boxed{\alpha_c^{\rm sd}(\beta_{\rm net},T)\;=\;\beta_{\rm net}\,[1-f_{\rm ret}(T)]-g_{\max}(\beta_{\rm net}).} \tag{9}$$

The dependence on $\beta_{\rm net}$ is real. At the right edge, the same $\beta_{\rm net}$ factor appears in retrieval and noise floor and cancels — that is why Ramsauer's bound is $\beta_{\rm net}$-independent. At the interior saddle the location $\varphi^*(\beta_{\rm net})$ itself shifts with $\beta_{\rm net}$, and the cancellation breaks. Figure 1 shows $\alpha_c^{\rm sd}(\beta_{\rm net},0)=\beta_{\rm net}-g_{\max}(\beta_{\rm net})$ together with Ramsauer's $\alpha_c^{\rm G}=1/2$. $\alpha_c^{\rm sd}(\beta_{\rm net},0)$ crosses Ramsauer's flat line at $\beta_{\rm net}\approx 0.7$: below this value the corrected capacity is more restrictive than Ramsauer's, above it more permissive. The endpoints are explicit: as $\beta_{\rm net}\to 0$, $\varphi^*\to\beta_{\rm net}$ and $g_{\max}\to\beta_{\rm net}^2/2$, giving the linear $\alpha_c^{\rm sd}(\beta_{\rm net},0)\to\beta_{\rm net}$; as $\beta_{\rm net}\to\infty$, $\varphi^*\to 1-1/(2\beta_{\rm net})$ and $g_{\max}\to\beta_{\rm net}-\frac{1}{2}(\ln\beta_{\rm net}+1)$, giving the logarithmic growth $\alpha_c^{\rm sd}(\beta_{\rm net},0)\to\frac{1}{2}(1+\ln\beta_{\rm net})$. The Monte Carlo simulations of Sections 3 and 4 fix $\beta_{\rm net}=1$ for the numerics, where $\alpha_c^{\rm sd}(1,0)\approx 0.6226$.

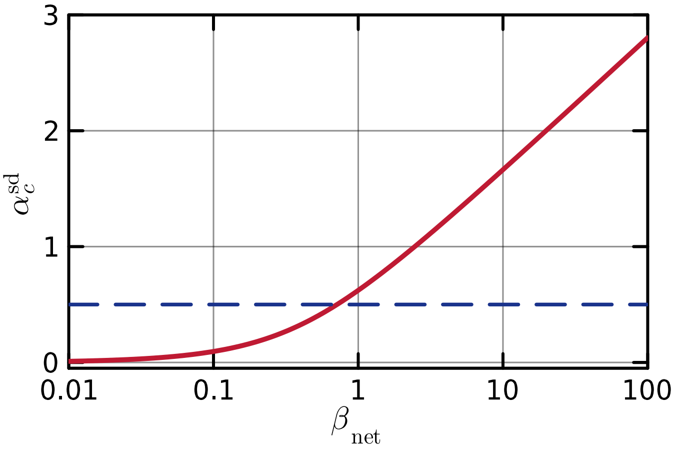


Figure 1: Zero-temperature capacity $\alpha_c^{\rm sd}(\beta_{\rm net}, 0)$ (9) of the LSE DAM as a function of the kernel sharpness $\beta_{\rm net}$ (red), together with Ramsauer's $\beta_{\rm net}$-independent bound $\alpha_c^{\rm G} = 1/2$ (5) (blue dashed).

**Cusp picture.** The saddle integral (8) is independent of $\boldsymbol{x}$: by isotropy of the random patterns, after averaging over the directions orthogonal to $\boldsymbol{\xi}^1$ only $\varphi_{1\mu}$ — not $\boldsymbol{x}$ itself — enters the spurious sum. The chain therefore competes inside the LSE logarithm between the retrieval term $\mathrm{e}^{\beta_{\rm net} N \varphi_1}$ and an $\boldsymbol{x}$-independent bulk $S \sim \mathrm{e}^{N[\alpha + g_{\max}(\beta_{\rm net})]}$. At large $N$,

$$\beta_{\rm net}^{-1} \ln[\mathrm{e}^{\beta_{\rm net} N \varphi_1} + S] \;\to\; \max(\varphi_1,\; \varphi_c(\alpha, \beta_{\rm net})), \qquad \varphi_c(\alpha, \beta_{\rm net}) \;=\; \frac{\alpha + g_{\max}(\beta_{\rm net})}{\beta_{\rm net}}, \tag{10}$$

and the leading-order free energy per spin is

$$\frac{F(\varphi_1; T, \alpha, \beta_{\rm net})}{N} \;=\; -\max(\varphi_1, \varphi_c) \;-\; \tfrac{T}{2} \ln(1 - \varphi_1^2). \tag{11}$$

The two branches meet in a cusp at $\varphi_1 = \varphi_c$. We call the right branch (slope $-1$) the *retrieval basin*: for $\varphi_1 > \varphi_c$ the retrieval term dominates the log-sum and the chain feels the linear pull of $\boldsymbol{\xi}^1$. We call the left branch (zero energy slope) the *saddle valley*: for $\varphi_1 < \varphi_c$ the bulk saddle dominates the log-sum, the energy in $\varphi_1$ is flat, and the sphere entropy $-(T/2)\ln(1 - \varphi_1^2)$ alone pulls the chain toward $\varphi_1 = 0$. Figure 2 plots (11) at $\alpha = 0.5$, $\beta_{\rm net} = 1$, $\varphi_c \approx 0.877$.

**Single-pattern spinodal.** The retrieval branch has a local minimum at $\phi_{\rm eq}(T)$ from (3), valid only while $\phi_{\rm eq}(T) > \varphi_c$. As $T$ rises, $\phi_{\rm eq}(T)$ drops; when it reaches $\varphi_c$ the local minimum disappears and the chain slides to $\varphi_1 = 0$. The condition $\phi_{\rm eq}(T) = \varphi_c$ defines the single-pattern spinodal

$$\alpha_c^{\rm sp}(\beta_{\rm net}, T) \;=\; \beta_{\rm net}\, \phi_{\rm eq}(T) - g_{\max}(\beta_{\rm net}), \qquad T_{\rm sp}(\alpha, \beta_{\rm net}) \;=\; \frac{\beta_{\rm net}^2 - (\alpha + g_{\max}(\beta_{\rm net}))^2}{\beta_{\rm net}\,(\alpha + g_{\max}(\beta_{\rm net}))}. \tag{12}$$

At $T = 0$, $\phi_{\rm eq}(0) = 1$, and the two curves (9) and (12) share the same endpoint $(\beta_{\rm net} - g_{\max}(\beta_{\rm net}), 0)$.

**Two boundaries, two protocols.** At $T > 0$ the spinodal $\alpha_c^{\rm sp}$ and the thermodynamic crossover $\alpha_c^{\rm sd}$ separate, with $\alpha_c^{\rm sd} < \alpha_c^{\rm sp}$ (since $\alpha_c^{\rm sp} - \alpha_c^{\rm sd} = -(T/2)\ln(1 - \phi_{\rm eq}^2) > 0$). In the strip $\alpha_c^{\rm sd}(\beta_{\rm net}, T) < \alpha < \alpha_c^{\rm sp}(\beta_{\rm net}, T)$ the basin still exists and lies above the cusp, but the bulk noise floor is lower in free energy. The escape barrier the chain must cross to leave the basin, $\Delta F_{\rm esc}/N = (\phi_{\rm eq} - \varphi_c) + (T/2)\ln[(1 - \phi_{\rm eq}^2)/(1 - \varphi_c^2)]$, depends only on $\phi_{\rm eq} - \varphi_c$ and vanishes at $\alpha_c^{\rm sp}$, not at $\alpha_c^{\rm sd}$. The Kramers escape time $\tau \sim \exp(N\, \Delta F_{\rm esc}/T)$ therefore grows exponentially with $N$ throughout $\alpha < \alpha_c^{\rm sp}$, regardless of which side of $\alpha_c^{\rm sd}$ we are on. Consequently:

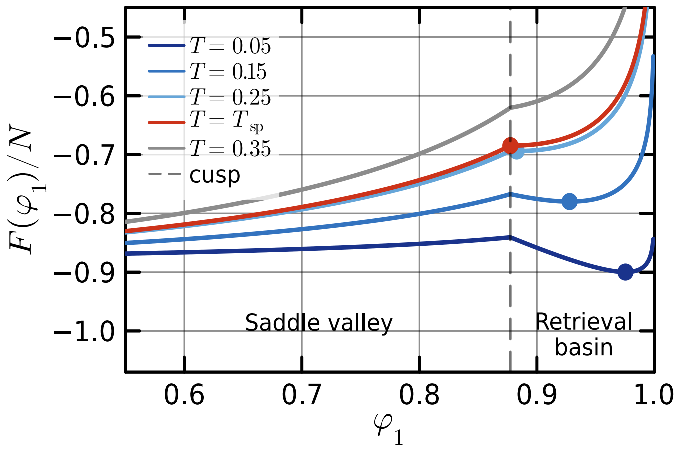


Figure 2: Leading-order free energy per spin $F(\varphi_1)/N$ from (11) at $\alpha = 0.5$, $\beta_{\rm net} = 1$, for five temperatures. The dashed vertical line marks the cusp $\varphi_c = \alpha + g_{\max} \approx 0.877$ separating the saddle valley on the left from the retrieval basin on the right. Dots: basin minima $\phi_{\rm eq}(T)$. At $T = T_{\rm sp}(0.5, 1) \approx 0.262$ the basin minimum reaches the cusp; for $T = 0.35 > T_{\rm sp}$ the basin no longer exists (grey curve).

- In the equilibrium (infinite-time) protocol, the chain Boltzmann-distributes between the basin and the noise floor; the empirical boundary is $\alpha_c^{\rm sd}$, the free-energy crossover.
- In the cold-initialised, finite-budget protocol — the practical retrieval scheme — the chain stays in the basin until its first Kramers escape. The asymptotic boundary at $N \to \infty$ is the spinodal $\alpha_c^{\rm sp}$, since $\tau \to \infty$ throughout $\alpha < \alpha_c^{\rm sp}$. At finite $N$ the empirical boundary is the Kramers contour $\tau(\alpha, T, N) = T_{\rm MC}$, which lies inside the strip $\alpha_c^{\rm sd} < \alpha < \alpha_c^{\rm sp}$ and drifts toward $\alpha_c^{\rm sp}$ as $N$ grows.

At $T = 0$ both curves coincide at $\beta_{\rm net} - g_{\max}(\beta_{\rm net})$ and the distinction disappears. Figure 3 confirms the picture at $N = 100$: $\log_{10}\langle\tau\rangle$ at five $\alpha$ approaching the $T = 0$ endpoint 0.6226 shows the basin surviving the full budget $10^6$ MC steps at low $T$ and dropping by several orders of magnitude over a $T$ window of width $\sim 0.01$ that slides toward $T = 0$ as $\alpha$ rises. A Kramers fit $\langle\tau\rangle \sim \tau_0\, {\rm e}^{c\Delta F/T}$ predicts that the product $T \log\langle\tau\rangle$ reads off the effective barrier $c\Delta F$; in our data this product traces a shallow $U$ in $T$ at each $\alpha$ — a finite high-$T$ prefactor on one side and a $T$-dependent barrier on the other ($\phi_{\rm eq}(T) - \varphi_c$ shrinks as $T$ rises). The minimum value of $T \log_{10}\langle\tau\rangle$ drops monotonically from $\approx 0.56$ at $\alpha = 0.50$ to $\approx 0.16$ at $\alpha = 0.58$ and below the measurement floor at $\alpha = 0.60$, consistent with the effective barrier vanishing as the spinodal is approached.

## 3 Monte Carlo schemes

We use three Metropolis variants on $\boldsymbol{x} \in \sqrt{N} \cdot S^{N-1}$, all at $\beta_{\rm net} = 1$. They differ in how they treat the $M = {\rm e}^{\alpha N}$ patterns inside the LSE log-sum.

**Direct.** All $M$ patterns are stored explicitly and the log-sum is evaluated over all of them at every step. Cost per step $\sim MN$. Direct MC is feasible up to $N \approx 25$ at $\alpha = 0.7$ ($M \approx 4 \times 10^7$); it is impossible at $N = 100$, $\alpha = 0.6$, where $M \approx 10^{26}$ exceeds available memory by twenty orders of magnitude.

**Truncated.** For each disorder draw we sample the $K \sim 10^4$ patterns whose overlap with $\boldsymbol{\xi}^1$ exceeds a cutoff $\varphi_{\rm keep}$. The cutoff is chosen so that the expected count above it equals the budget

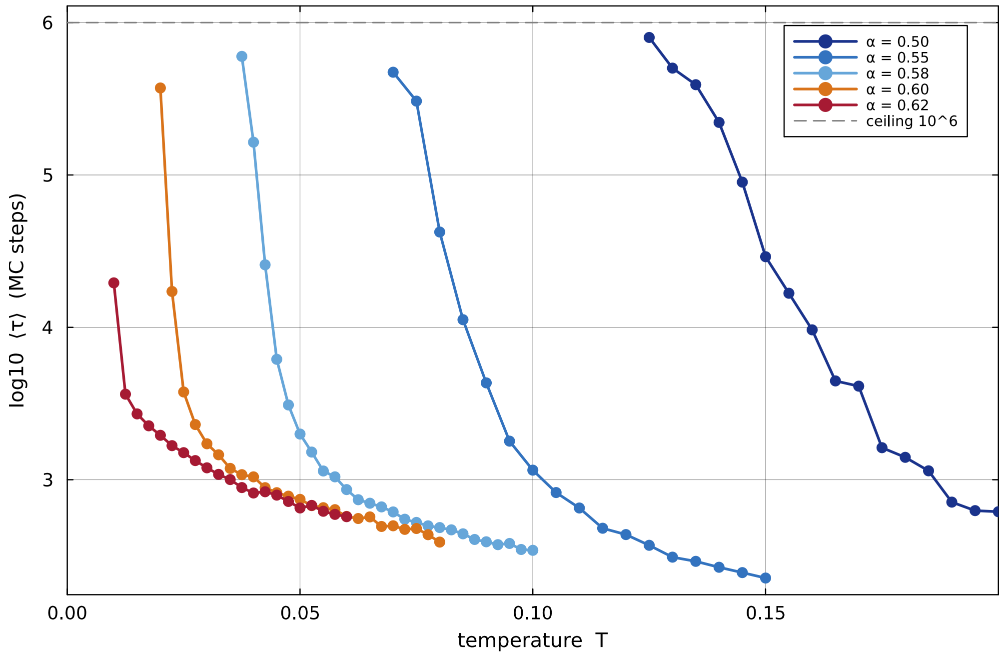


Figure 3: Mean basin escape time $\langle\tau\rangle$ versus $T$ at $N = 100$ from N-dim hybrid MC, 32 disorder samples per cell, $\beta_{\text{net}} = 1$. Five values of $\alpha$ approaching $\alpha_c^{\text{sd}}(1, 0) \approx 0.6226$. Cells in which no disorder escaped within $10^6$ MC steps are not shown; the grey dashed line marks that budget ceiling. The escape window narrows and slides toward $T = 0$ as $\alpha$ rises.

$K$. The remaining $M - K$ patterns below the cutoff enter the LSE log-sum through an analytic constant $\ln C_{\text{bulk}} = \alpha N + \ln \int_{-1}^{\varphi_{\text{keep}}} f(\varphi)\, \mathrm{e}^{N\varphi}\, \mathrm{d}\varphi$. The retained set is fixed once per disorder. At $N = 25$ truncated and direct produce the same basin boundary across the full $\alpha$ range we measure, so the discarded patterns are genuinely outside the basin physics rather than silently dropped. Truncated is the workhorse for the wide heatmap in Section 4.

**Hybrid.** Same decomposition as truncated, with two changes: the cutoff is set at the cusp position $\varphi_c(\alpha, \beta_{\text{net}})$ from (10) rather than chosen by budget, and the bulk integral is replaced by its saddle-point value $M\, \mathrm{e}^{N g_{\max}(\beta_{\text{net}})}$ — an $\boldsymbol{x}$-independent constant inside the log-sum. Patterns above $\varphi_c$ are drawn from the spherical density (2) restricted to $\varphi > \varphi_c$. In the rescaled variable $y = (1 + \varphi)/2 \in [0, 1]$, the spherical density becomes a symmetric Beta:

$$y \sim \mathrm{Beta}\Big(\tfrac{N-1}{2}, \tfrac{N-1}{2}\Big) \ \text{ truncated to } \ y \in \big[(1 + \varphi_c)/2,\, 1\big], \qquad \varphi_{1\mu} = 2y - 1. \tag{13}$$

Each retained pattern is then assembled from its $\varphi_{1\mu}$ and a uniform direction $u_\mu$ on the $(N-2)$-sphere perpendicular to $\boldsymbol{\xi}^1$. At the loads and dimension studied here ($N = 100$, $\alpha \in [0.5, 0.62]$), the expected count of overlaps above $\varphi_c$ is below unity for every $\alpha$, so the algorithm typically retains only $\boldsymbol{\xi}^1$ explicitly and treats every other pattern through the bulk constant. This is a faithful sampler of the leading-order free energy (11): below the cusp the bulk constant inside the log-sum collapses $\beta_{\text{net}}^{-1} \ln[\cdot]$ to the constant $\varphi_c$ — the noise-floor branch — but the chain still feels the sphere geometry through the entropy term $-(T/2)\ln(1 - \varphi_1^2)$, which is what drives the slide to $\varphi_1 = 0$.

**Why local refresh fails.** A faster scheme — sometimes called "smart" — refreshes the retained set every $N_{\text{refresh}}$ steps from a cone around the current chain position, keeping only the patterns whose overlap with $\boldsymbol{x}$ is currently large. The refresh breaks the quenched-disorder assumption that underlies (3)–(9). As $\boldsymbol{x}$ wanders, new patterns are minted at every refresh, so the cumulative

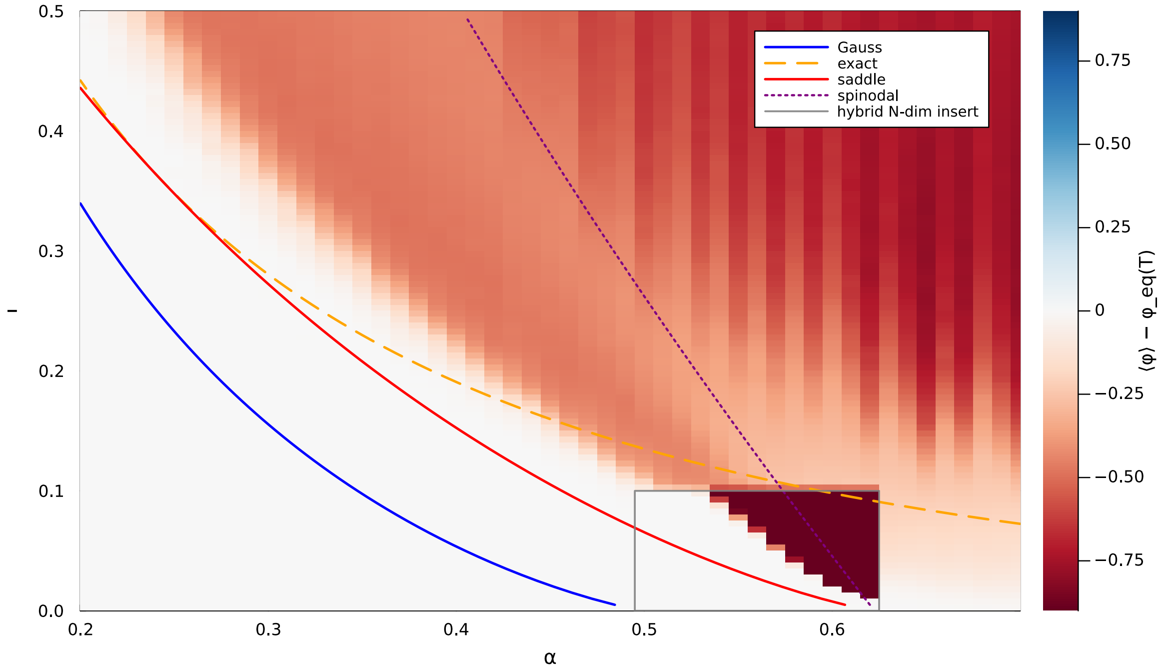


Figure 4: $\langle\varphi_1\rangle - \phi_{\rm eq}(T)$ from Monte Carlo over the $(\alpha, T)$ plane at $\beta_{\rm net} = 1$. Background: truncated MC at varying $N$ ($\alpha \in [0.20, 0.70]$, $M \approx 4.4 \times 10^6$). Grey rectangle: N-dim hybrid MC at $N = 100$ on a fine $T$ grid. Curves: $\alpha_c^{\rm G}$ (5) (blue solid, Gaussian + support edge); $\alpha_c^{\rm bd}$ (6) (orange dashed, exact density + support edge); $\alpha_c^{\rm sd}$ (9) (red solid, thermodynamic F-crossover); $\alpha_c^{\rm sp}$ (12) (purple dotted, single-pattern spinodal). $\alpha_c^{\rm sd}$ and $\alpha_c^{\rm sp}$ share the endpoint at $T = 0$ where the basin disappears; the empirical line at finite $N$ sits between them, on the Kramers contour, drifting toward $\alpha_c^{\rm sp}$ as $N$ grows.

set of patterns seen along a single trajectory exceeds $M$, and the chain effectively faces a denser pattern set than the model defines. In the retrieval basin the spurious-recruitment rate alone destroys retrieval — for reasons unrelated to (9). Truncated and hybrid both fix the retained set once per disorder and avoid the issue.

# 4 Results

Figure 4 shows $\langle\varphi_1\rangle - \phi_{\rm eq}(T)$ over the $(\alpha, T)$ plane at $\beta_{\rm net} = 1$. The bulk of the plane, $\alpha \in [0.20, 0.70]$, uses truncated MC at varying $N$ (the load $M = e^{\alpha N}$ is held fixed at $M \approx 4.4 \times 10^6$ by ramping $N$ with $\alpha$). A fine $T$-grid hybrid MC at $N = 100$ fills in the rectangle $\alpha \in [0.5, 0.62]$, $T \in [0, 0.10]$.

The basin region (white) is bounded by a sharp dark-red transition that pinches at $\alpha \approx 0.6226$ at $T = 0$. Ramsauer's $\alpha_c^{\rm G} = 1/2$ from (5) and the exact-edge curve $\alpha_c^{\rm bd}$ from (6) both sit visibly off the empirical boundary at low $T$. The thermodynamic crossover $\alpha_c^{\rm sd}$ from (9) (red) and the spinodal $\alpha_c^{\rm sp}$ from (12) (purple) bound a strip $\alpha_c^{\rm sd} < \alpha < \alpha_c^{\rm sp}$ at $T > 0$, sharing the single endpoint at $T = 0$. The empirical line at $N = 100$ sits inside this strip, on the Kramers contour $\tau(\alpha, T, N) = T_{\rm MC}$ where the chain escapes within the simulation budget. As $N$ grows the contour drifts toward $\alpha_c^{\rm sp}$ — the cold-initialised asymptotic boundary discussed in §2.

# 5 Discussion

The Ramsauer threshold $\alpha_c = 1/2$ is a Gaussian artefact. The Gaussian density misses the interior saddle that the exact spherical density places at $\varphi^*(\beta_{\rm net})$ in the partition-sum integrand; restoring it gives a finite $\beta_{\rm net}$-dependent capacity $\alpha_c^{\rm sd}(\beta_{\rm net}, 0) = \alpha_c^{\rm sp}(\beta_{\rm net}, 0) = \beta_{\rm net} - g_{\rm max}(\beta_{\rm net})$ — the load at which the cusp $\varphi_c = (\alpha + g_{\max})/\beta_{\text{net}}$ reaches the unit sphere and the retrieval basin disappears. The capacity sits below Ramsauer's bound at moderate $\beta_{\text{net}}$, crosses it at $\beta_{\text{net}} \approx 0.7$, and grows logarithmically beyond. The $\beta_{\text{net}}$-independence of Ramsauer's $1/2$ is itself the signature of the right-edge replacement: at the edge the $\beta_{\text{net}}^{-1}$ prefactor of $H$ and the $\beta_{\text{net}}$ inside the exponent cancel, while at the interior saddle the cancellation breaks through the $\beta_{\text{net}}$-dependence of $\varphi^*$.

At $T > 0$ the saddle line $\alpha_c^{\text{sd}}$ and the spinodal $\alpha_c^{\text{sp}}$ separate. $\alpha_c^{\text{sd}}$ is the thermodynamic boundary at which the basin and the saddle valley exchange order in free energy; $\alpha_c^{\text{sp}}$ is the geometric boundary at which the cusp catches the basin minimum and the retrieval branch disappears. For a cold-initialised chain the cusp is the only escape route, with barrier $(\phi_{\text{eq}} - \varphi_c) + (T/2)\ln[(1-\phi_{\text{eq}}^2)/(1-\varphi_c^2)]$ vanishing only at $\alpha_c^{\text{sp}}$. The Kramers escape time grows exponentially in $N$ throughout $\alpha < \alpha_c^{\text{sp}}$, irrespective of which side of $\alpha_c^{\text{sd}}$ the chain sits on; at $N \to \infty$ the empirical boundary is therefore the spinodal. At finite $N$ it is the Kramers contour inside the strip $\alpha_c^{\text{sd}} < \alpha < \alpha_c^{\text{sp}}$, which is what our hybrid MC at $N = 100$ measures.

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